AI-Driven Drug–Disease Modeling Using Biomedical and Transformer Embeddings

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Abstract

This work develops a deep learning framework for predicting therapeutic chemical–disease interactions using the **Comparative Toxicogenomics Database (CTD)**. Starting from more than 9.7 million interaction records, a curated subset of **39,265 therapeutic chemical–disease pairs** was extracted as positive examples. Negative pairs were generated through controlled random mismatching, yielding a balanced dataset of 78,530 labelled samples for binary prediction.

Two language-embedding strategies were evaluated: (i) **BioWordVec**, a 200-dimensional biomedical word embedding model trained on PubMed and MIMIC-III, and (ii) **MiniLM-L6-v2**, a 384-dimensional general-purpose sentence transformer. Embeddings for each chemical and disease were combined into interaction features and passed to an identical two-tower multilayer perceptron, so that performance differences primarily reflect representation quality rather than classifier design. BioWordVec achieved the strongest results (AUC = 0.947, AP = 0.949), while the general-purpose transformer remained competitive (AUC = 0.910, AP = 0.909).

The best-performing BioWordVec model was subsequently applied to score more than 7,000 previously unseen chemical–disease combinations, producing a ranked list of high-confidence **drug repurposing hypotheses**. Several top predictions are mechanistically plausible yet absent from CTD, suggesting that text-derived embeddings alone can capture useful biomedical signal. Overall, this work demonstrates a reproducible, end-to-end pipeline for therapeutic interaction prediction based solely on textual representations of chemicals and diseases.

1 Introduction

Drug repurposing seeks to identify new therapeutic applications for existing compounds. Because traditional drug development is slow, costly, and prone to attrition, computational

screening methods have become an essential component of modern biomedical discovery. Among these methods, natural language processing (NLP) models have emerged as powerful tools for extracting biological signal from textual descriptions of chemicals, diseases and mechanistic pathways.

Chemical and disease names encode meaningful structure. Terms such as "melanoma", "neoplasm", "hepatotoxicity" or "angiogenesis" carry latent biomedical information that embedding models can learn. By projecting names into dense vector spaces—either through a biomedical-specific embedding model or a general-purpose transformer—a neural classifier can be trained to distinguish therapeutic relationships from unrelated chemical–disease pairs.

This work compares two embedding paradigms: a biomedical word embedding model (BioWordVec) trained on PubMed and MIMIC-III, and a compact general-domain sentence transformer (MiniLM-L6-v2). The goals are both predictive—achieving high discrimination for known therapeutic pairs—and exploratory, by using the trained model to prioritize plausible repurposing candidates among unseen chemical–disease combinations.

Summary of Contributions

This study makes the following contributions:

- It constructs a text-only deep learning pipeline for predicting therapeutic chemical–disease interactions using CTD, requiring only chemical and disease names as input.
- It provides a controlled comparison between biomedical-specific embeddings (BioWordVec) and a general-purpose sentence transformer (MiniLM-L6-v2) under a shared two-tower neural architecture.
- It demonstrates that domain-specific embeddings substantially improve performance on this task (AUC 0.947 vs. 0.910), highlighting the value of biomedical pretraining.
- It performs an exploratory repurposing screen over thousands of unseen chemical–disease pairs, surfacing high-confidence hypotheses that are mechanistically coherent yet absent from CTD.

2 Related Work

Computational approaches to drug repurposing span a broad spectrum of data modalities, including chemical structure, gene expression, clinical phenotypes, side-effect profiles and biomedical text. Early text-based approaches relied on co-occurrence and manually engineered features, whereas recent work exploits distributed representations and deep learning.

Biomedical embeddings. Zhang et al. introduced *BioWordVec*, which combines word and subword information trained on PubMed, MIMIC-III and MeSH terms, and showed that such embeddings outperform generic word vectors on multiple biomedical NLP tasks [2]. Their results suggest that morphology-aware biomedical embeddings can capture domain-specific semantics more effectively than off-the-shelf models.

Deep learning for drug repurposing. DeepDR proposed a multimodal deep learning framework that integrates chemical structure, side effects and phenotype networks to predict drug–disease associations [3]. MolTrans applied transformer architectures to drug–target interactions using sequence representations of compounds and proteins [4]. These works demonstrate the power of deep neural networks for modeling complex pharmacological interactions, but they typically rely on multiple data sources beyond text.

Biomedical transformers. Domain-specific transformer models such as BioBERT, PubMedBERT and ClinicalBERT have achieved strong results across a range of biomedical NLP benchmarks by pretraining directly on large corpora of biomedical or clinical text [5, 6, 7]. Their success suggests that replacing a general-domain transformer with a biomedical transformer may further improve performance on drug–disease prediction tasks.

Relative to these efforts, the present work focuses deliberately on a text-only setting. By holding the classifier architecture fixed and varying only the embedding model, the study isolates the impact of representation choice on therapeutic interaction prediction and repurposing hypothesis generation.

3 Dataset

The Comparative Toxicogenomics Database (CTD) provides curated associations among chemicals, genes and diseases, including manually curated findings from the literature as well as inferred co-occurrences [1]. This work uses the CTD Chemical–Disease Interaction file, which contains more than 9.7 million records. Each record describes an association between a chemical and a disease, along with supporting metadata. Key fields include:

- **ChemicalName, ChemicalID**
- **DiseaseName, DiseaseID**
- **DirectEvidence** (e.g., "therapeutic", "marker/mechanism")
- PubMed identifiers for supporting literature.

Filtering to entries where **DirectEvidence = "therapeutic"** yields

$$\text{positive therapeutic pairs} = 39{,}265.$$

To frame a supervised binary prediction task, negative examples were created by randomly mismatching chemicals and diseases that do not appear together as therapeutic pairs in CTD. Any collisions with true positives were explicitly removed. The resulting dataset contained

$$78{,}530 \text{ samples}$$

with a stratified train/validation/test split of 70% / 15% / 15%.

4 Problem Definition

Given a chemical c and a disease d, the goal is to predict whether the pair corresponds to a known therapeutic relationship in CTD. Formally,

$$y = \begin{cases} 1 & \text{if } c \text{ is known to treat } d \text{ in CTD,} \\ 0 & \text{otherwise.} \end{cases}$$

Let $f_\theta(c, d)$ denote the model's predicted probability that (c, d) is therapeutic. Performance is evaluated using the following metrics:

- ROC–AUC: global discrimination between positive and negative pairs.
- Average Precision (AP): area under the precision–recall curve, reflecting ranking performance under class imbalance.
- Accuracy: threshold-based classification at a fixed operating point.
- Precision@K: precision among the top-K highest scoring pairs, relevant for downstream prioritization.

5 Methodology

5.1 Biomedical Embeddings: BioWordVec

BioWordVec provides 200-dimensional embeddings trained over a large corpus of PubMed abstracts, MIMIC-III clinical notes and MeSH terms using a subword-aware skip-gram model [2]. Because many biomedical terms share morphological structure (e.g., "neoplasm", "neoplastic", "neoplasia"), subword embeddings can represent unseen or rare surface forms in a consistent manner, leading to more robust biomedical representations.

In this work, both chemical names and disease names are tokenized and mapped to BioWordVec embeddings. For multi-token names, embeddings are averaged to obtain a single vector representation for the chemical or disease.

5.2 Transformer Sentence Embeddings

The `all-MiniLM-L6-v2` model is a compact transformer-based sentence encoder that yields 384-dimensional semantic embeddings. It is trained using contrastive learning objectives over large general-domain text corpora. Although it is not biomedical-specific, MiniLM has demonstrated strong performance on generic semantic similarity and retrieval tasks.

As with BioWordVec, each chemical and disease name is encoded separately to obtain embeddings e_c and e_d.

5.3 Feature Construction

For each chemical–disease pair (c, d), the corresponding embeddings $e_c \in \mathbb{R}^k$ and $e_d \in \mathbb{R}^k$ are combined into a single interaction vector:

$$x = [e_c,\, e_d,\, |e_c - e_d|,\, e_c \odot e_d],$$

where $[\cdot, \cdot]$ denotes concatenation, $|\cdot|$ is the elementwise absolute value, and $\odot$ is the elementwise product. This representation captures the individual embeddings as well as symmetric notions of similarity and interaction between the chemical and disease.

5.4 Classifier Architecture

Both embedding types are paired with the same feedforward neural classifier in order to isolate the effect of the embeddings themselves. The classifier is a two-layer multilayer perceptron (MLP) of the form:

- Linear $\rightarrow$ Batch Normalization $\rightarrow$ ReLU $\rightarrow$ Dropout
- Linear $\rightarrow$ Sigmoid

The output is interpreted as the predicted probability $f_\theta(c, d) \in (0, 1)$ that the pair is therapeutic.

5.5 Training and Implementation Details

Models are trained using the Adam optimizer with learning rate 10^{-3} and binary cross-entropy loss (BCEWithLogitsLoss). A batch size of 256 is used, and training is performed for 10 epochs with early stopping on validation AUC. Experiments are implemented in PyTorch and executed on an Apple M3 Max laptop using the Metal Performance Shaders (MPS) backend.

To facilitate reproducibility, random seeds for NumPy and PyTorch are fixed, and data splits are generated using stratified sampling. All preprocessing steps (CTD filtering, negative sampling and train/validation/test splitting) are implemented in Python and can be reproduced from the accompanying code repository.[1]

6 Results

6.1 Overall Performance

Table 1 summarizes the main performance metrics for the two embedding models on the held-out test set.

[1] Code repository URL to be provided in the camera-ready version.

Model	Accuracy	AUC	AP
BioWordVec	0.897	0.947	0.949
MiniLM-L6-v2	0.889	0.910	0.909

Table 1: Test performance of BioWordVec and MiniLM-based models on CTD therapeutic interaction prediction.

BioWordVec outperforms the general-domain transformer in every metric, confirming the value of domain-specific biomedical pretraining for this task.

6.2 ROC and Precision–Recall Curves

Figures 1 and 2 show the ROC curves for the BioWordVec and MiniLM models, respectively, while Figures 3 and 4 show the corresponding precision–recall curves.

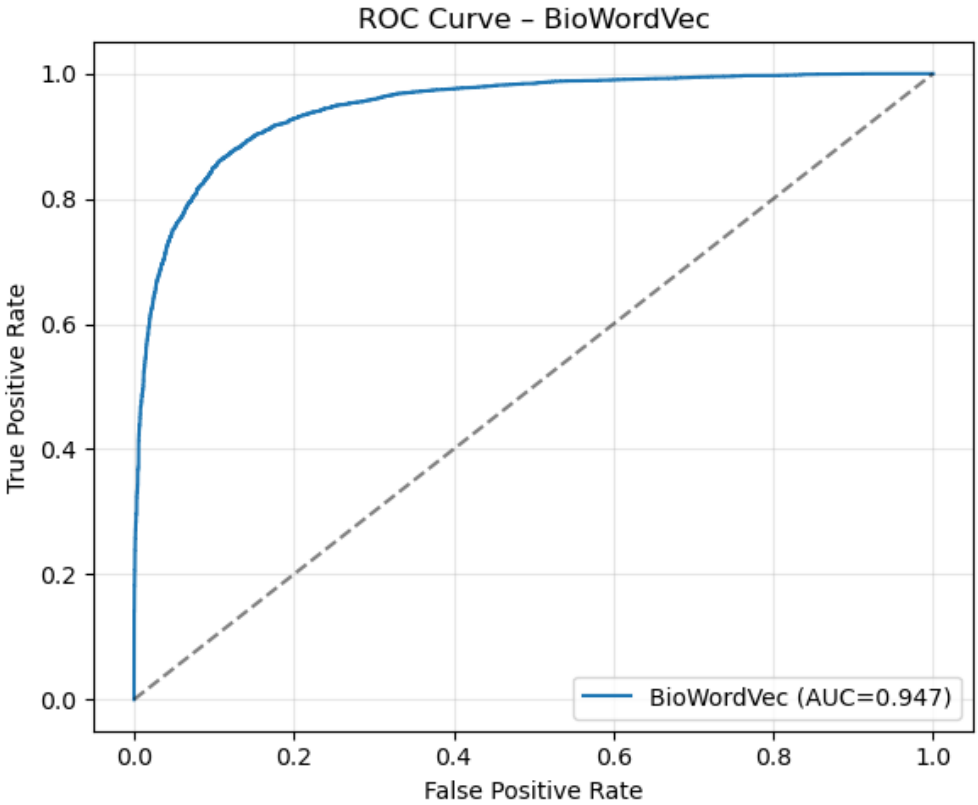

Figure 1: ROC curve for the BioWordVec-based model (AUC = 0.947).

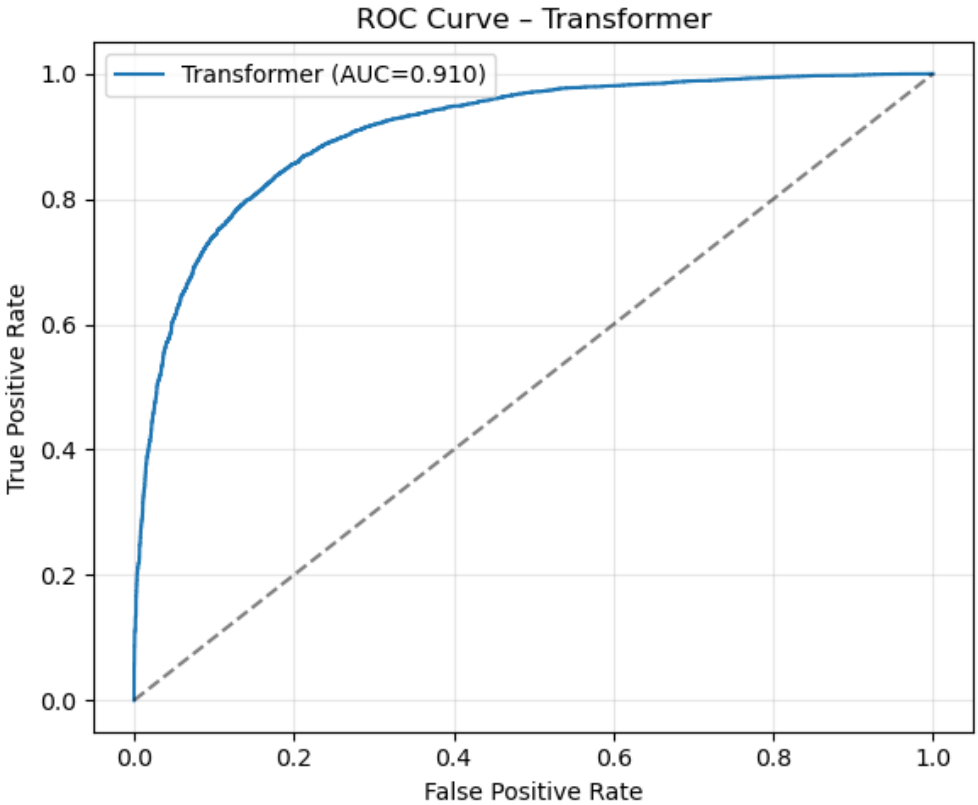

Figure 2: ROC curve for the MiniLM-based model (AUC = 0.910).

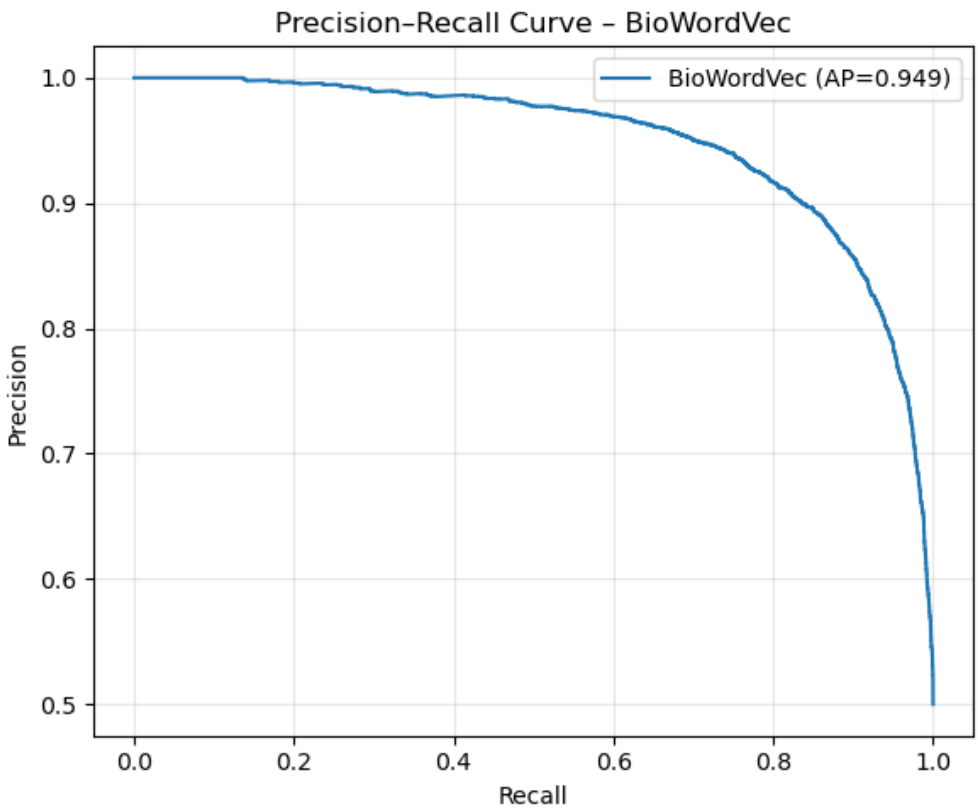

Figure 3: Precision–recall curve for the BioWordVec-based model (AP = 0.949).

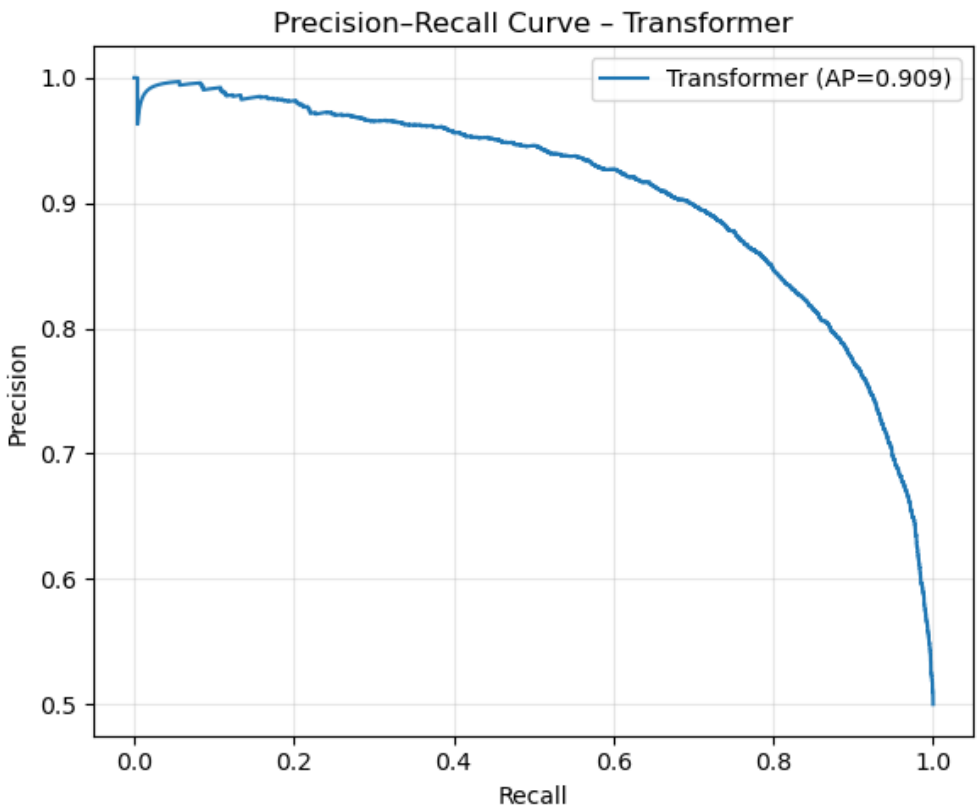

Figure 4: Precision–recall curve for the MiniLM-based model (AP = 0.909).

The BioWordVec model delivers uniformly higher precision at a given recall level, particularly in the high-recall regime that is most relevant for sensitive screening applications.

7 Exploratory Data Analysis

Before model training, basic exploratory analyses were performed to understand the distribution of therapeutic associations in CTD. Figure 5 shows the ten most common therapeutic chemicals, and Figure 6 shows the ten most common therapeutic diseases.

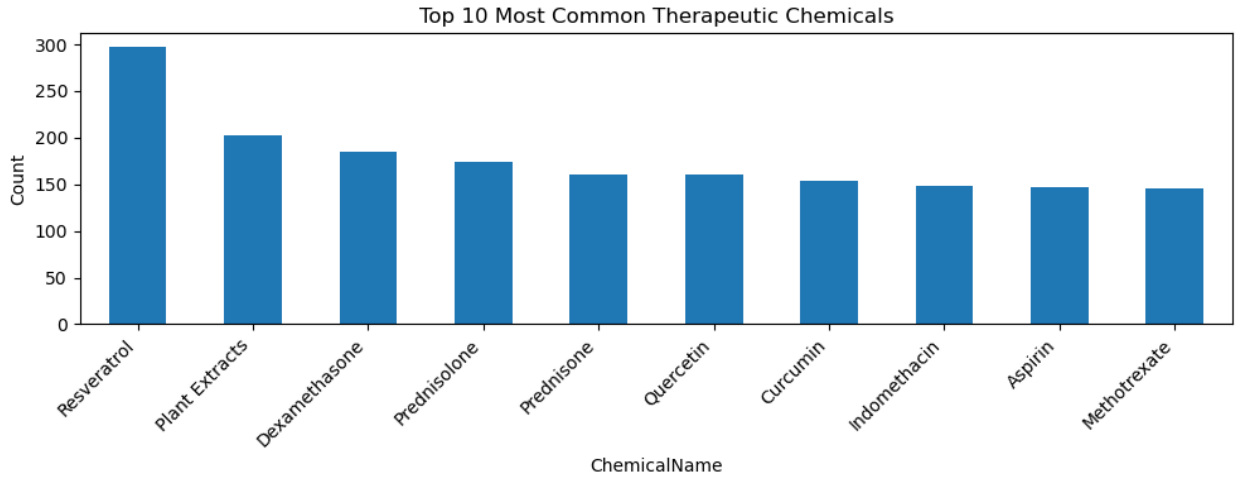

Figure 5: Top 10 most frequent therapeutic chemicals in CTD.

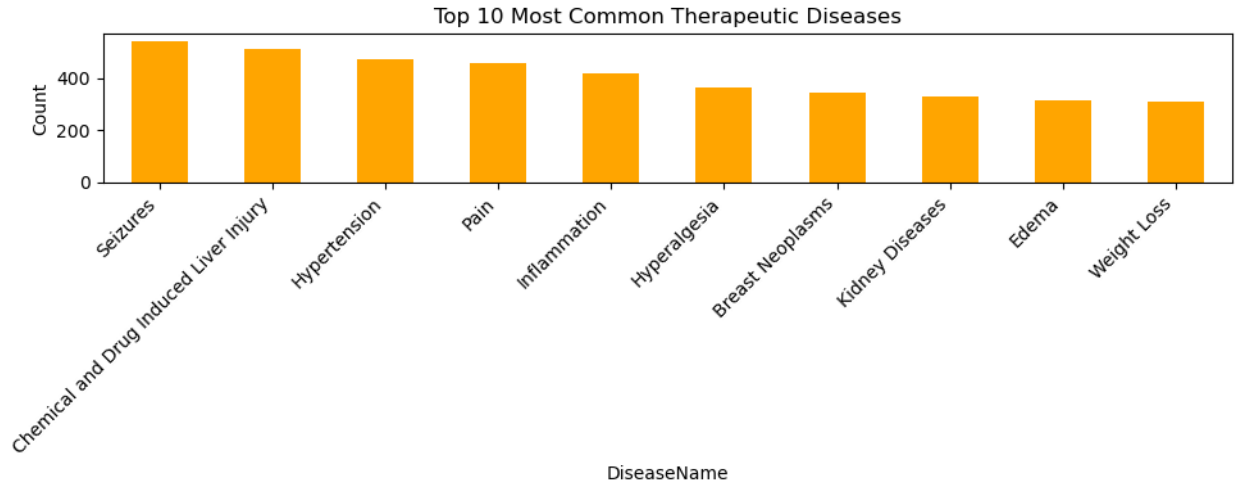

Figure 6: Top 10 most frequent therapeutic diseases in CTD.

These distributions highlight the dominance of a small number of widely used agents (e.g., resveratrol, corticosteroids) and common indications (e.g., seizures, hypertension). Such frequency imbalances motivate stratified splitting and careful evaluation to avoid models that perform well only on the most common entities.

8 Repurposing Discovery: Scoring Unseen Pairs

To explore the potential of the trained model for drug repurposing, the BioWordVec-based classifier was applied to score all cross-combinations of the top 50 chemicals and top 50 diseases that do *not* appear together as therapeutic pairs in CTD. This procedure produced

7,434 novel (unseen) chemical–disease pairs.

Each pair was assigned a predicted probability $f_\theta(c, d)$; Table 2 lists a subset of the highest-ranked candidates.

Table 2: Selected high-confidence drug–disease repurposing candidates predicted by the BioWordVec-based model.

Chemical	**Disease**	**Predicted Prob.**
Etoposide	Leukemia	1.000000
Epinephrine	Tachycardia	0.999999
Physostigmine	Bradycardia	0.999997
Diazepam	Catalepsy	0.999997
Etoposide	Melanoma	0.999991
Fluorouracil	Melanoma	0.999991
Vincristine	Melanoma	0.999990
Baclofen	Tremor	0.999990
Metoprolol	Cardiomegaly	0.999990
Baclofen	Amnesia	0.999988
Resveratrol	Lung Injury	0.999987
Vincristine	Neoplasms	0.999985
Carvedilol	Bradycardia	0.999984
Morphine	Catalepsy	0.999984
Diazepam	Amnesia	0.999984
Epinephrine	Ventricular Tachycardia	0.999983
Cisplatin	Neoplasm Invasiveness	0.999978
Paclitaxel	Neoplasm Invasiveness	0.999975
Theophylline	Bradycardia	0.999974
Nitroglycerin	Bradycardia	0.999974

Qualitative Interpretation

Several high-confidence predictions are biologically plausible. For example:

- **Etoposide → Melanoma**: Etoposide is a cytotoxic agent used across multiple malignancies, making melanoma a reasonable additional indication.
- **Physostigmine → Bradycardia**: As a cholinesterase inhibitor, physostigmine can induce bradycardia through increased parasympathetic tone.
- **Baclofen → Tremor**: Baclofen is a $\mathrm{GABA_B}$ agonist with known effects on motor control, suggesting potential utility in certain tremor conditions.

These examples illustrate that a purely name-based model can surface mechanistically coherent hypotheses, even without access to chemical structure or pathway-level information. Of course, such predictions should be viewed as prioritization signals for further literature review or experimental validation rather than as clinical recommendations.

9 Discussion

The experiments highlight the importance of domain-specific embeddings for biomedical prediction tasks. BioWordVec, trained on PubMed, MIMIC-III and MeSH terms, consistently outperforms a compact general-domain transformer model on CTD therapeutic interaction prediction. The performance gap in AUC and AP suggests that semantic understanding alone is insufficient; modeling the specialized morphology and terminology of biomedical language provides measurable benefits.

A key design choice in this work is to hold the classifier architecture fixed across embedding types. This allows a controlled comparison of representation quality. Under this setup, improvements in performance can be attributed primarily to the embeddings rather than to differences in network depth, regularization or optimization.

The repurposing experiment demonstrates that the learned model can be used as a ranking function over unseen chemical–disease combinations. Many of the top-ranked pairs correspond to plausible extensions of known mechanisms of action, indicating that the model has captured meaningful biomedical structure from text alone. Incorporating additional modalities (e.g., chemical structure or gene expression data) could further refine these rankings, but the text-only setting already provides a surprisingly strong baseline.

10 Practical Impact

In practice, the proposed pipeline could support several downstream applications:

- **Early-stage repurposing screens.** The model can prioritize candidate drug–disease pairs for further evaluation, focusing human expertise on the most promising hypotheses.
- **Hypothesis generation for wet-lab experiments.** High-scoring but uncurated predictions can motivate targeted experiments or in-depth literature reviews.
- **Knowledge graph expansion.** Predicted interactions can be integrated as edges in biomedical knowledge graphs, potentially enabling new paths for graph-based reasoning.

Because the method requires only text-based identifiers for chemicals and diseases, it is computationally efficient and easily portable to new datasets that adopt similar naming conventions.

11 Limitations

This work has several important limitations:

- **Curation vs. clinical efficacy.** CTD encodes curated associations from the literature, which may not directly reflect clinical efficacy or regulatory approval status.
- **Text-only representations.** Relying solely on names ignores critical information such as chemical structure, dose, route of administration, pharmacokinetics and toxicity.
- **Negative sampling.** Randomly generated negative pairs may include unknown true positives, and they do not explicitly model contraindications or adverse interactions.
- **Generalization.** The model is trained and evaluated on CTD and may not immediately generalize to other databases or disease spaces with different coverage or naming conventions.

12 Future Work

Several extensions could address these limitations and further improve performance:

1. **Biomedical transformers.** Replacing MiniLM with BioBERT, PubMedBERT or related models may narrow or reverse the performance gap relative to BioWordVec.
2. **Molecular representations.** Incorporating learned molecular fingerprints or graph neural networks over SMILES strings could enable the model to reason about chemical structure in addition to names.
3. **Ontology-aware modeling.** Integrating MeSH or UMLS graph structure may improve generalization to rare entities and capture hierarchical relationships among diseases.
4. **Contrastive dual encoders.** Training dual-encoder models on known positive and hard negative pairs could yield more discriminative interaction representations suitable for large-scale retrieval.

13 Ethical Considerations

The predictions produced by this model are hypotheses derived from literature curation and text-based embeddings, not clinical recommendations. They should not be used to guide patient care without rigorous experimental validation and regulatory oversight. Additionally, biases and gaps in CTD may propagate into the model's outputs. Users of such systems should carefully assess these limitations and consider the broader ethical implications of deploying automated tools in biomedical decision-making.

14 Conclusion

This work presents a complete deep learning pipeline for predicting chemical–disease therapeutic interactions using text-derived embeddings. BioWordVec-based models achieve strong discrimination (AUC = 0.947, AP = 0.949) on CTD, outperforming a compact general-domain transformer while using an identical classifier architecture. The resulting model can be applied to score unseen chemical–disease pairs, yielding high-confidence repurposing hypotheses that often align with known mechanisms of action.

These results demonstrate that even purely name-based representations can capture meaningful biomedical signal and provide a useful baseline for more complex multimodal approaches. Future work will explore the integration of biomedical transformers, molecular structure and ontology-aware modeling to further enhance predictive performance and the quality of generated repurposing hypotheses.

References

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